Biologia, Bratislava, **61**/Suppl. 19: S300—S304, 2006
Section Botany
DOI: 10.2478/s11756-006-0177-7

Soil water regime of agricultural field and forest ecosystems

Vlasta Štekauerová[1], Viliam Nagy[1] & Dana Kotorová[2]

[1]*Institute of Hydrology of the Slovak Academy of Sciences, Račianska 75, SK–83102 Bratislava, Slovakia; tel.: ++421-2-44259404, fax: ++421-2-44259404, e-mail: stekauer@uh.savba.sk, nagy@uh.savba.sk*
[2]*Slovak Agricultural Research Centre – Institute of Agroecology, Michalovce, Špitálska 1273, SK–07101 Michalovce, Slovakia; e-mail: kotorova@minet.sk*

Abstract: The unsaturated zone of soil is one of the most important and complicated parts considering the water movement in the hydrologic cycle. Water transfer through its upper and lower boundary directly influences the amount of water in this zone. The depth of groundwater table usually delimits the lower boundary. The soil surface with or without plant canopy is the upper boundary. The soil surface reacts directly on meteorological conditions primary through evapotranspiration. It is determining the inflow of precipitation into deeper layers of a soil profile. Both physical and hydrophysical properties of the upper soil layer are changed under the extreme meteorological conditions. In case of such conditions the water can flow along preferential pathways down to the groundwater without filling up the soil matrix. Changes of physical and hydrophysical characteristics of the soil surface layer or of the root zone depend on the vegetation type, too. The water storage of 0–30 cm and 30–60 cm soil layers was calculated from monitored data of soil water contents in two different ecosystems, and the calculated water storages were compared with the integral water contents that are related to hydrolimits (field capacity, point of decreased availability and wilting point) in 1999 and 2000. It should be noted that it was considerably dryer in 2000 than in 1999, and during the vegetation period, it was also warmer in 2000 than in 1999. The higher air temperatures during the vegetation season and the lower cumulative rainfall in 2000 comparing to 1999 resulted in a decrease in the integral water contents in the root zone.

Key words: hydraulic conductivity, hydrolimit, precipitation, soil water retention curve, water storage

Introduction

The water regime describes the integral changes of water content in the unsaturated soil zone during a certain period. Long term monitoring of the water regime gives the possibility for classifying the soil types, and for fulfilling expectations of various research subjects, such as ecology, hydrology or agronomy. The water regime of the soil which determines the productivity, depends on inflow and outflow of water in the unsaturated soil zone.

In lowlands the inflow into the unsaturated soil zone consists of two main components infiltration of precipitation and upwards movement of capillary water from the groundwater. These two components highly regulate the water content of this zone. The water outflow from the unsaturated zone can be due to the evaporation of the bare soil surface, the transpiration of the vegetation, or drainage to the groundwater (Merta et al., 2006; Buchtele et al., 2006). The soil surface can react directly on the meteorological and climate conditions (Stehlová, 2004).

The water regime of the unsaturated soil zone is influenced by different factors of which the most important are: the height of the groundwater table related to relief of the field, the soil hydrophysical characteristics, the climatic conditions, the vegetation and the anthropogenic impact (Čermák & Kučera, 1987). It is known that soil as vital entity of the biosphere is far from a homogeneous medium (Mikulec, 2004; Halabuk, 2006; Farkas et al., 2006).

In case of the water regime evaluation it is important to take into consideration the available amount of water from the soil for the plant in different ecosystems. Hydrolimit is the soil water content, reached under a certain condition (Kutílek, 1978). Mostly its value does not have an exact physical significance. However, the value of the hydrolimit is different for plant species and varieties. Hydrolimits are useful to calculate the water balance in a soil profile, for instance, to define the quantity of water available for or used by the plants. The method using hydrolimits to estimate seasonal water regime was successfully used by Šútor et al. (2002).

There are three hydrolimits which can be used for assessing the water requirements of plants. These are the field capacity (FC), the point of decreased availability (PDA), and the wilting point (WP). FC is defined as the soil water content within the water potential interval of pF = 2.3 and pF = 2.7. PDA is defined as the

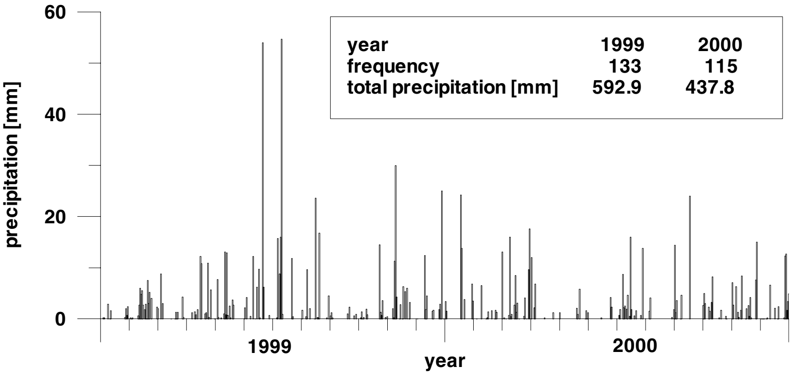

Fig. 1. Daily precipitation measured in Gabčíkovo in 1999 and 2000.

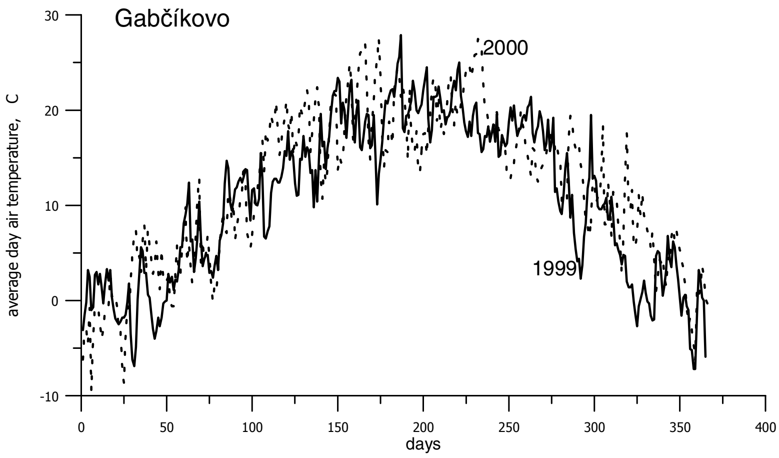

Fig. 2. Average daily air temperature measured at the meteorological station in Gabčíkovo in 1999 and 2000.

soil water content within the water potential interval of pF = 3.1 and pF = 3.5. WP is defined at the water potential of pF = 4.18. At this soil water content plants suffer from water deficit or drought.

Water retention curves of the soil layers are used to obtain the hydrolimits (Štekauerová et al., 2002; Orfánus, 2005). In case the soil profile is not vertically homogeneous and consists of significantly different layers, water retention curves of the respective layers have to be determined. Soil scientists and ecologists are also used to the concept of studying the available water amount for plants during different vegetation periods (e.g. Rajkai, 1983).

The aim of this study is to calculate the integral water content of the 0–30 cm and 30–60 cm soil layers from their monitored soil water contents, and to define the amount of plant available water in soils of cultivated fields and of a forest ecosystem in 1999 and 2000.

Locality of the study area

Regions around the Danube – Žitný ostrov (left side of the river) are bordered by a natural water way called Little Danube. The region is crossed by channels which regulate the groundwater level. Along the river Danube the heterogeneity describes best the soils of the area which leads to heavily layered soil profiles. Some parts of the region are under intensive agricultural use and some parts are covered by a forest ecosystem. The monitored agricultural field is located in Báč and the forest ecosystem is in Bodíky. Crops in the agricultural field are generally maize, cereals, rape, and sunflower. The dominant tree species in the forest ecosystem is the Canadian poplar which is mainly used by the timber industry. The forest of the investigated area the forest is about 20 years old. Both places are situated on lowland.

The soil profiles in both places are mutually very simple. The texture of the soil surface is mainly loam and it gradually turns to sandy-loam and sand downwards. The sand is passing into a gravelly subsoil in the deeper layers at both places. The sand layer begins approximately at a depth of 120 cm in Bodíky and at 90–100 cm in Báč. The soil profile in Báč is considerably more layered than in Bodíky.

The saturated hydraulic conductivity values measured by a Guelph permeameter at depths of 30 cm were: 7.63×10^{-6} m s^{-1} (Bodíky), 1.81×10^{-7} m s^{-1} (Báč).

The level of the groundwater table is fluctuating only in the gravel subsoil and is therefore not influencing the water regime of the 0–60 cm soil layer. The meteorological characteristics such as the average daily temperature and daily precipitation have been measured in the Gabčíkovo meteorological station (Figs 1, 2).

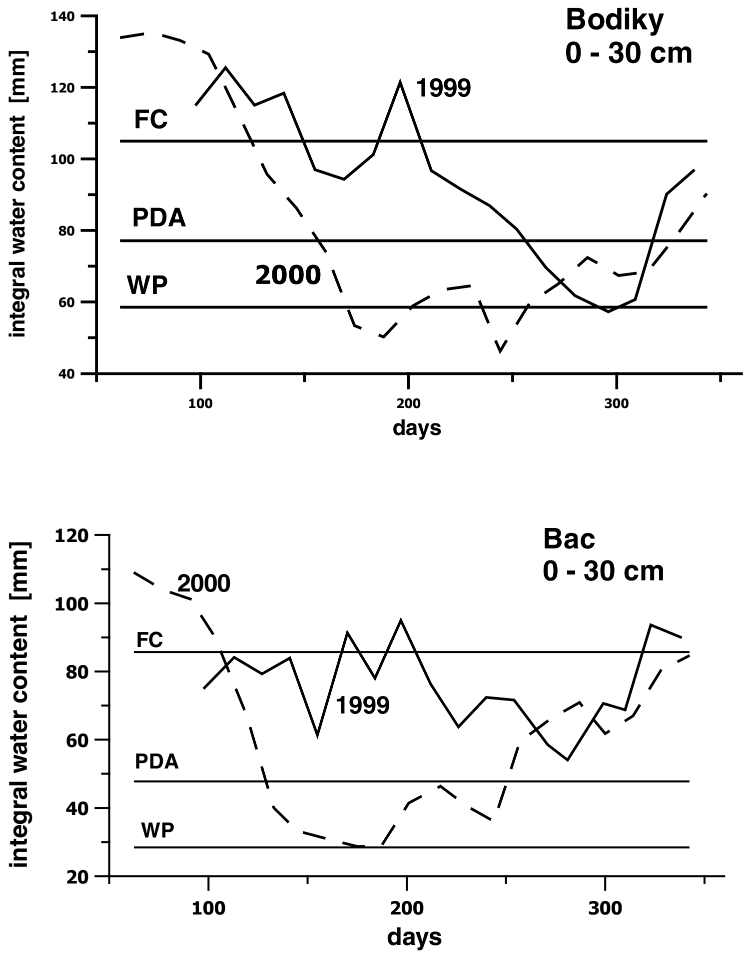

Fig. 3. Integral water contents and hydrolimits (FC: field capacity, PDA: point of decreased availability, WP: wilting point) of the 0–30 cm soil layer in Bodíky and Báč in 1999 and 2000.

Material and methods

Soil water contents were measured from the soil surface to the groundwater level at 10 cm intervals by neutron probe every two weeks (in winter every month) during 1999 and 2000. The calibration curves for the neutron probe were made in various seasons in order to give higher precision as producer of the neutron probe advises. Integral water contents of soil layers were calculated from the soil water contents monitored by the neutron probe.

Drying branches of the soil water retention curves were measured on the soil samples under laboratory conditions by overpressure apparatus (The soil Moisture Equipment, Santa Barbara) and they were approximated using the relation of van Genuchten (Van Genuchten, 1980). The approximated soil water retention curves were used for establishing the hydrolimits as field capacity, point of decreased availability for plants and wilting point. The soil profile was not homogeneous vertically and consisted of layers with different composition and properties. For that reason we determined the water retention curves of each layer of the soil profile (Štekauerová & Nagy, 2002). Water supply to plants in the different ecosystems was estimated from the integral water contents in relation to the distinguished hydrolimits.

Results and discussion

Figures 3 and 4 show the seasonal changes of the integral water content in the 0–30 cm and 30–60 cm soil layers. It is obvious from the figures that roots can take up water from the 0–60 cm soil layer.

The meteorological station in Gabčíkovo recorded 592.9 mm of rainfall in 1999 and 437.8 mm in 2000. The year 2000 was considerably dryer (about 17.4%) than 1999, and the air temperature was higher in the vegetation period.

The groundwater table courses were almost identical at both study sites in 1999 and 2000. However, the water regime of the 0–60 cm soil layer was not influenced by the groundwater table (Štekauerová & Nagy, 2003).

The water storage of the 0–60 cm soil layer was filled up by precipitation. The effect of precipitation on the integral water content of soils is visible in figures 3 and 4. The integral water content of the 0–30 cm soil layer is less in 2000 at both localities. The same is valid for the 30–60 cm layer with relatively lower water contents at Báč. It can also be seen in the figures 3 and

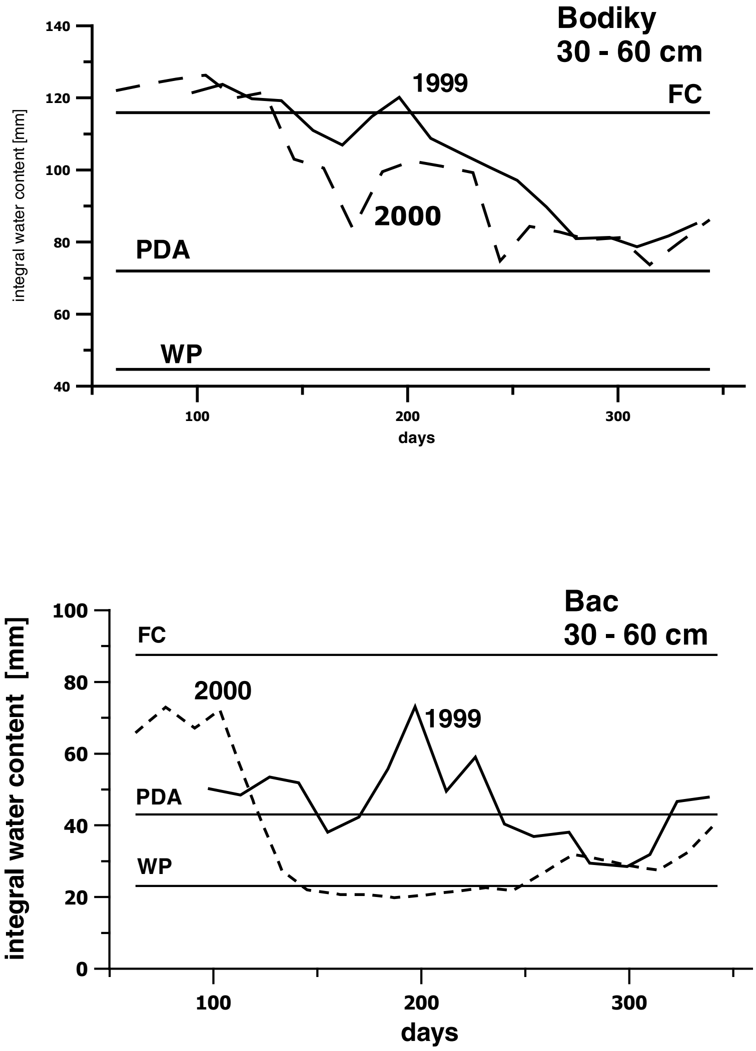

Fig. 4. Integral water contents hydrolimits (FC: field capacity, PDA: point of decreased availability, WP: wilting point) of the 30–60 cm soil layer in Bodíky and Báč in 1999 and 2000.

4 that the upper layers of the forest soil contain more water than the tilled (and non-irrigated) soil in both years. The ground water table was fluctuating almost identically in the two consecutive years.

The integral water content related to that of the decreased availability (PDA) in the 0–30 cm soil layer at Bodiky and Báč was lower for most of the year in 2000. Even it was lower than the integral water amount of the wilting point in Bodíky.

The integral water quantity of the 30–60cm soil layer was higher than the integral water equivalent of the PDA in the two years. However, the integral water amount of this deeper soil layer was close to the wilting point integral water content in the year 2000 at Báč.

Conclusions

The points of the soil water retention curve drying branches were measured for several soil profile layers at localities of Bodíky and Báč, and they were approximated using the relation of van Genuchten. The hydrolimits for the soil water potential $pF_{FC} = 2.3$ (field capacity), $pF_{PDA} = 3.3$ (point of decreased availability), and for soil water potential $pF_{WP} = 4.18$ (wilting point) were used for assessing the water requirements of plants. The integral water contents of 0–30 cm and 30–60 cm soil layers were calculated from monitored soil water contents by neutron probe at both localities during the years 1999 and 2000.

As it was considerably dryer in 2000 than in 1999, and during the vegetation period, it was also warmer in 2000 than in 1999, the integral water contents in the root zone were lower in 2000 than in 1999 (Figs 3 and 4). The integral water contents were often below the wilting point in the tilled soil (Báč), and sometimes in the forest ecosystem (Bodíky). It could be concluded that the forest ecosystem is less susceptible to decrease in the integral water contents in the root zone as the tilled soil.

Acknowledgements

The authors acknowledge the financial support from the projects VEGA 2/5018//5, APVT 51-044802, and APVT 51-019804.

References

Buchtele, J., Buchtelová, M. & Tesař, M. 2006. Role of vegetation in the variability of water regimes in the Šumava Mts forest. Biologia, Bratislava **61(Supl. 19):** S246–S250.

Čermák, J. & Kučera, J. 2006. Transpiration of mature stands of spruce (*Picea abies* (L.) Karst.) as estimated by the tree-trunk heat balance method, pp. 311–317. In: Proc. Symp. Forest hydrology and watershed management. IAHS Publ. No. 167, Wallingford.

Farkas, Cs., Gyuricza, Cs. & Birkas, M. 2006. Seasonal changes of hydraulic properties of a Chromic Luvisol under different soil management. Biologia, Bratislava **61(Supl. 19):** S344–S348.

Halabuk, A. 2006. Influence of different vegetation types on saturated hydraulic conductivity in alluvial topsoils. Biologia, Bratislava **61(Supl. 19):** S334–S337.

Kutilek, M. 1978. Soil science for water management. SNTL-ALFA, Prague. (In Czech)

Merta, M., Seidler, Ch. & Fjodorowa, T. 2006. Estimation of evaporation components in agricultural crops. Biologia, Bratislava **61(Supl. 19):** S280–S283.

Mikulec, V. 2004. Impact of saturated hydraulic conductivity of soils on numerical simulation of soil water movement. In: Pollution and water resources. Columbia University Seminar Proceedings, Vol. XXXV, SAS, Bratislava, pp. 193–202.

Orfánus, T. 2005. Spatial assessment of soil drought indicators at regional scale: hydrolimits and soil water storage capacity in Záhorská nížina Lowland. Hydrol. Hydromech. **53:** 164–176.

Rajkai, K. 1983. Ecological application of soil hydrophysical properties (in Hungarian). PhD thesis. RISSAC HAS, Budapest, 137 pp.

Stehlová, K. 2004. Assessment of change of soil water content in the soil profile at locality Bodíky in comparison with a forecast of its potential change in the time horizon 2010, 2030, and 2075, pp. 203–215. In: Halasi-Kun, G.J. (ed.), Pollution and water resources. Columbia University Seminar Proceedings, Vol. XXXV, SAS, Bratislava.

Štekauerová, V. & Nagy, V. 2003. Estimation of soil zone aeration water regime at Žitný ostrov localities. Acta Hydrologica Slovaca, **4:** 45–73. (In Slovak)

Štekauerová, V. & Nagy, V. 2002. Influence of climate conditions on security necessary water for vegetation in various ecosystems. In: Halasi-Kun, G.J. (ed.), Pollution and water resources. Columbia University Seminar Proceedings, Vol. XXXIII-XXXIV, HAS, Budapest, 324–337.

Štekauerová, V., Skalová, J. & Šútor, J. 2002. Using of pedotransfer functions for assessment of hydrolimits. Rostlinná Výroba **48:** 407–412.

Šútor, J., Štekauerová, V. & Majerčák, J. 2002. Climate changes and water regime of soil aeration zone in Slovakia lowlands. I. Analysis of an impact of expected changes in average monthly temperatures. Acta Hydrologica Slovaca, **3:** 143–154 (In Slovak).

Van Genuchten, M.Th. 1980. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. Soil Sci. Soc. Am. J. **44:** 987–996.